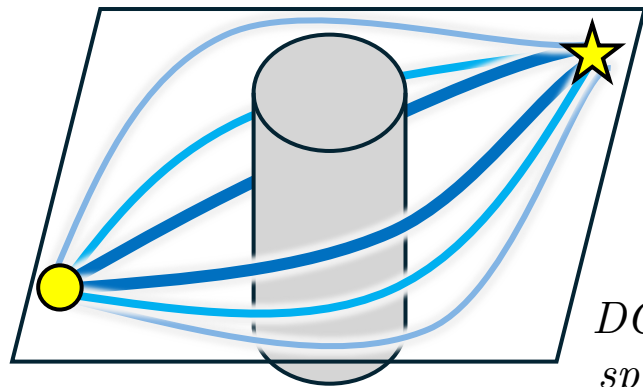


A. Sampling based Predictive Control with DCBF guided Sampling

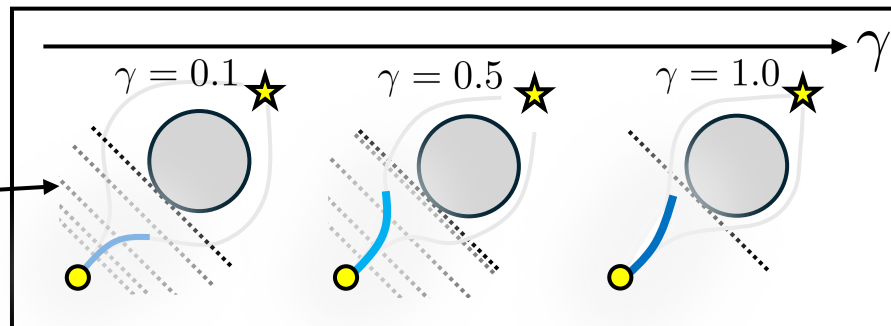
Safe action prior in distribution



DCBF safety specification

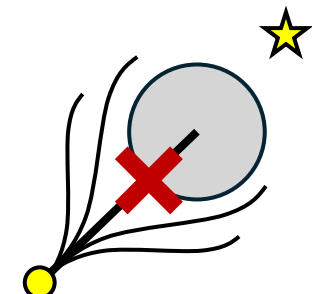
MPPI-DCBF (Ours)

- Certified safe averaged control
- Rapid safe demonstrations



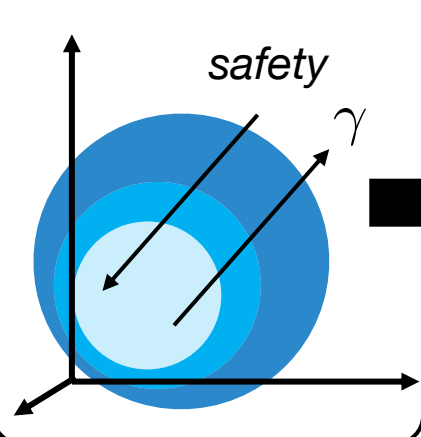
Prior works (MPPI safety)

- Unsafe cost-weighted averaged control
- Sacrifice optimality



B. Offline Generative Control Policy Training

Pretrained CFM



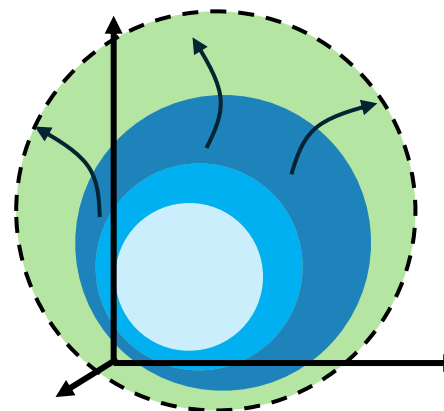
Uncertainty-guided Self-generation

Safe Flow Expansion

CFM Update

Safety Verifier

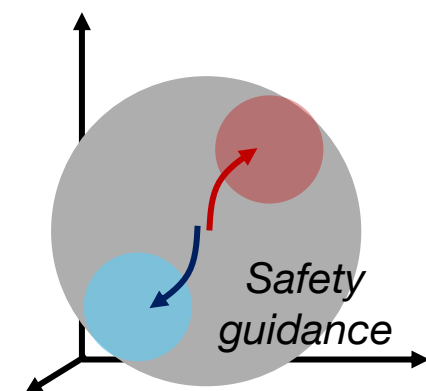
Safe Flow Expanded CFM (Ours)



Expanded safe generable set

- Safety tunable generative policy
- Discover new safe behaviors

Prior works (Guided CFM)



Fixed generable set

- Reward guided generative policy
- Limited OOD coverage

C. Online Trajectory Generation in OOD

